



An experimental study of n-dodecane and the development of an improved kinetic model

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ABSTRACT

This work presents new fundamental combustion experimental datasets and propose an improved kinetic model for n-dodecane, a widely used surrogate component for jet fuels and diesel. In the experiments, ignition delay times for n-dodecane, were measured over low-to-high temperature range by combining a heated rapid compression machine (RCM) and a heated shock tube (ST). The measurements cover a wide range of temperatures (621–1320 K), pressures (8, 15 bar) and equivalence ratios (0.5–1.5). Flow reactor oxidation was also carried out over temperature range of 600–1100 K and at equivalence ratios of 0.5, 1.0 and 1.5 under atmospheric pressure to provide species evolution profiles of O₂, CO, CO₂, CH₄, C₂H₄ etc. These datasets were used to develop a mechanism for n-dodecane. New rate coefficients were adopted from recent literature to improve the performance of the model, leading to a model composed of 737 species and 3629 reactions. The present experimental data, as well as a wide range of datasets in literature, were used to evaluate the performance of the model. An overall good agreement of the predictions with the experimental results was observed. This model is anticipated to facilitate the development of kinetic models for jet fuel surrogate.

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1. Introduction

It is well established that accurate simulation using kinetic mechanism of surrogate fuels contributes to the understanding of the interaction between chemical reaction pathway and flow [1–3] for complex real fuels, and is helpful for identifying and controlling the pollutant formation, e.g., soot [4–7]. Obviously, the maturity of kinetic mechanisms of pure surrogate components determine the performance of the surrogate mechanism. N-dodecane is a long-chain linear alkane widely existing in real transport fuels. Due to the appropriate carbon number and relevance to jet fuels, it is often used as the representative n-alkane in multi-component jet fuel surrogates and sometimes as a single-component surrogate for jet fuels [8–13]. It also serves as a component in some proposed diesel surrogates [14,15]. Therefore, a good understanding of the combustion chemistry of n-dodecane is important to the design of modern engines and the optimization of combustion strategies.

The combustion chemistry of n-dodecane has been a focus of interest in the past decades, and there have been extensive researches focusing on experimental characterization and ki-

netic modeling of this fuel. Previous experimental studies for n-dodecane have been performed in shock tubes (ST), flow reactor, jet-stirred reactor (JSR) and flames, including the measurements of ignition delay time (IDT), speciation of pyrolysis and oxidation, and laminar flame speed. The previous experimental studies are illustrated in Table 1. The kinetic targets provided in these fundamental combustion experiments are indispensable for the development of kinetic mechanisms. Note that the IDT of n-dodecane, a crucial global kinetic parameter for the validation of kinetic model, was seldom studied over low-to-intermediate temperature range using RCM. Although conventional aero-engines usually work at high temperatures, the knowledge about the low temperature chemistry of n-dodecane is significant in the context of using jet fuels in diesel or HCCI type engines. Furthermore, the available IDT measurements for n-dodecane were all preformed in air. However, the promising low temperature combustion (LTC) technology relies on very lean or extremely diluted combustion [16], it is of great significance to characterize the autoignition of fuels under diluted conditions.

Parts of the above fundamental combustion experiment databases have been used as kinetic targets and helped to pave the way for the development, validation, refinement and reduction of n-dodecane combustion kinetic models. In addition, some kinetic

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Table 1
Previous fundamental combustion experiments for n-dodecane.

Reactor	T (K)	Pressures	Equivalence ratio	Type of data	Ref.
ST	727–1422	15–34 atm	0.5, 1	IDTs	[17]
Aerosol ST	1050–1350	5.8–6.7 atm	0.4–0.57	IDT	[18]
ST	786–1396	14, 40 atm	0.5, 1.0	IDTs	[19]
ST	715–1288	14–70 atm	1	IDTs	[20]
ST	1300–1600	Near 2 atm	1	Species time history	[21]
ST	709–792	Around 10 atm	1.0	CO time history, 1st IDTs	[22]
ST	867–1739	19–74 atm	0.46–2.05	Oxidation species	[23]
Pyrolytic reactor	1200–1600	600 torr	–	Pyrolysis species	[24]
Flow reactor	750–1430	0.0066–1 atm	–	Pyrolysis species	[25]
Pressurized flow reactor	1000–1300	1 atm	0.79–1.56/∞	Species	[26]
Pressurized flow reactor	550–830	0.8106 MPa	0.23	Oxidation species	[27]
JSR	550–1150	10 bar	0.5, 1.0, 2.0	Oxidation species	[28]
JSR	773–1073	1 atm	–	Pyrolysis species	[29]
Counterflow configuration	400, 470	1 atm	0.7–1.4	Laminar flame speed	[30]
Counterflow configuration	400	1, 2, 3 atm	0.7–1.3	Laminar flame speed	[31]
Counterflow configuration	403	1 atm	0.7–1.5	Laminar flame speed	[32]

studies also contributed to kinetic mechanism development for n-dodecane. For instance, Wang et al. [33] investigated the initiation mechanism and product distributions of pyrolysis and combustion of n-dodecane; Zhao et al. conducted a computational study on the unimolecular decomposition of n-dodecane [24]. Based on these previous works, several detailed mechanisms for n-dodecane have been published, including the models proposed by Ranzi et al. (called Ranzi's model) [34], Biet et al. (called Biet's model) [35], Wang et al. (called JetSurF model series) [26,36,37], Westbrook et al. (called LLNL model) [38,39], Mzé-Ahmed's et al. (called Mzé-Ahmed's model) [28], as well as a very recently developed model by Zeng et al. (called Zeng's model) [25]. In addition, several reduced or optimized models [7,14,40–42] were derived from these detailed models. Detailed mechanism with thousands of species and elementary reactions is very useful in many circumstances and is a prerequisite to derive reduced models, and thus determines the performance of the derived reduced mechanism. Therefore, of particular interest in this work is detailed models or optimized models, while less attentions are paid to reduced models. The status of the development of detailed mechanism is briefly reviewed as follows. Ranzi et al. developed a semi-empirical and lumped mechanism for heavy linear alkanes including n-dodecane for both pyrolysis and oxidation spanning low to high temperature ranges, and later updated and improved it in the following studies [43,44]. However, these models failed to simulate the experimental data at intermediate and high temperatures, [43] which demonstrated the need for further improvement. Biet's model was developed using computer-aided generation software EXGAS [35]. This model was validated against the flow reactor oxidation experiment and the results showed that the mechanism needed further improvement. The n-dodecane sub-mechanism in JetSurF model was taken from the n-dodecane mechanism proposed by You et al. [45]. Banerjee et al. optimized the sub-mechanism of n-dodecane in JetSurF 1.0 model based on their experimental datasets, leading to the release of the latest version of this model, i.e., Optimized JetSurF 1.0 model [26]. To our knowledge, this model mainly focused on the high temperature chemistry and only very limited reactions were used to describe the low temperature reaction pathway. Hence, its performance in predicting low temperature oxidation needs to be further evaluated. Westbrook et al. [38] proposed a detailed model for n-alkanes from C8 to C16, and later Sarathy et al. [39] updated and extended this model to include the reaction schemes of 2-methylalkanes from C7–C20, leading to the latest LLNL model. This model was used to simulate the experimental results in many studies [20–23,26–28,46], and the observations implied the necessity to revisiting the kinetics to improve its predictions. Starting from the LLNL model, Narayanaswamy et al.

obtained a skeletal model by using mechanism reduction and then revisited the rate constants for some reaction classes, leading to the release of Narayanaswamy's model [42]. This model was extensively validated against literature data, but the observed discrepancy suggests that further improvements are required to improve its performance. Another improved n-dodecane model derived from LLNL model was derived by Cai et al. [41], who updated the model with the optimized rate rules. Mzé-Ahmed's et al. [28] proposed a detailed mechanism to describe combustion of n-dodecane over the entire temperature range. The model was compared with their JSR oxidation speciation profiles and literature data, and was found to overestimate the IDTs at low-to-intermediate temperatures. Very recently, Zeng et al. [25] developed a detailed model for n-dodecane and compared it with their pyrolysis data, as well as extensive datasets in literature. Although good agreements were observed between simulations and experimental data, the low temperature reaction scheme needs further evaluation.

In summary, the experimental studies on the autoignition characteristics of n-dodecane are inadequate, leaving a gap under low-to-intermediate temperature range and diluted conditions. On the other hand, discrepancies were observed when using the existing kinetic models to simulate the experimental databases, implying further improvements are needed for these models. Furthermore, some of these experimental data were available after the publication of the above-mentioned kinetic models, and thereby those models were not validated against all existing data. Therefore, there are still some experimental databases that can be utilized for further model evaluation and improvement. These research insufficiencies motivate the interest of the present study. This study aims to improve the knowledge of n-dodecane oxidation chemistry by (1) providing a reliable IDT data set over low-to-high temperature ranges by combining a heated RCM and a heated ST; (2) providing speciation information of oxidation in a flow reactor and (3) proposing an improved kinetic model of good performance over wide conditions.

2. Methods

2.1. Ignition delay time measurements

The test fuel used in this study is n-dodecane with a purity of 99% provided by MACKLIN. The IDT measurements were conducted in the heated RCM and ST at Shanghai Jiao Tong University (SJTU). The details of the two facilities can be found in our previous studies [47–52], and will not be repeated here. Since n-dodecane is a heavy hydrocarbon with high boiling point, a preheating temperature of 115 °C was used to ensure the complete vaporization of the

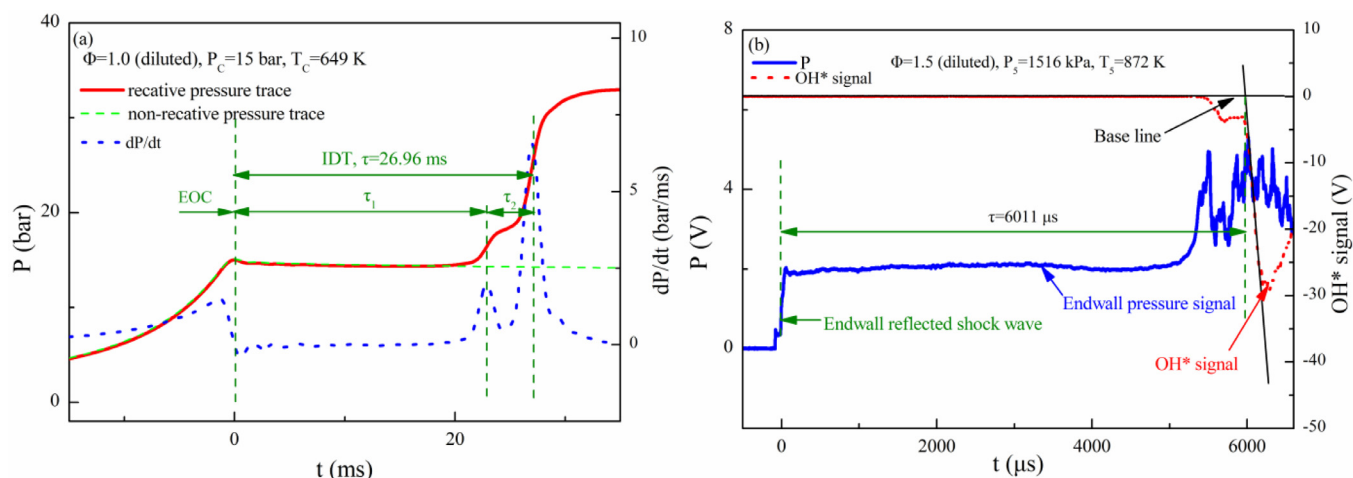


Fig. 1. Definition of IDTs in measurements: (a) RCM measurement and (b) ST measurement.

Table 2
Test conditions used in IDT measurements for n-dodecane.

Equivalent ratio	x_{fuel} (mol%)	x_{O_2} (mol%)	x_{N_2} (mol%)	Dilution ratio ($x_{N_2}/(x_{N_2}+x_{O_2})$)	Pressure (bar)
0.5	0.56	20.88	78.55	79.0%	8/15
0.5	0.28	10.47	89.25	89.5%	8/15
1.0	0.56	10.44	88.99	89.5%	8/15
1.5	0.84	10.41	88.74	89.5%	8/15
1.5	0.56	6.96	92.48	93.0%	8/15

fuel. The monitoring results showed that the vaporization rate was higher than 99% throughout the experiments. Before the first experimental run, the prepared mixture was maintained at the preset temperature for at least 3 h to ensure homogeneity. For RCM measurements, a minimum of 3 parallel runs was carried out at each test condition to ensure the repeatability. A non-reactive run was conducted under each test condition by replacing the oxygen with equal nitrogen to mimic the facility effect, including the heat loss and potential reactions during the compression stroke. The IDT in RCM measurements was defined as the time from end of compression to the maximum pressure rise rate, as shown in Fig. 1. For ST measurements, pure He was usually used as driver gas. However, tailored He/N₂ mixtures were used as driver gas to extend the effective experimental time when the IDT of the test mixture was estimated to be longer than 2 ms. The IDT in ST measurements was defined as the time from the arrival of the reflected shock wave at the endwall to the onset of ignition, as shown in Fig. 1. The test conditions adopted in this study are listed in Table 2. The uncertainty of RCM measurement is estimated to be less than $\pm 10\%$ [48,53], while maximum uncertainty of ST measurement is estimated to be $\pm 20\%$ based on an Arrhenius-type correlation using the root-sum-squares method [54].

Figure 2 shows a comparison between the present IDTs and the literature dataset measured under similar conditions. Generally, the two datasets show good agreement over the intermediate-to-high temperature range. The data of Vasu et al. basically falls within the experimental error bars, and the temperature window of negative temperature coefficient (NTC) behavior follows well with the newly measured data. The good agreement demonstrates the reliability of the present dataset.

2.2. Flow reactor oxidation

In this study, the oxidation speciation profiles of n-dodecane were measured in the plug flow reactor at SJTU. Since the detailed

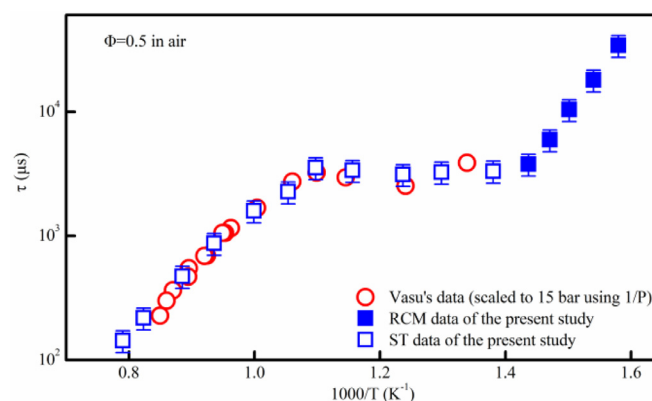


Fig. 2. Comparison between current data and literature data [17] for autoignition of n-dodecane.

Table 3
Test conditions of n-dodecane flow reactor oxidation experiments.

Equivalence ratio	Mole fraction (%)		
	x_{fuel}	x_{O_2}	x_{Ar}
0.5	0.30	11.10	88.60
1.0	0.30	5.55	94.15
1.5	0.30	3.70	96.00

description of this apparatus can be found elsewhere [55–57], it is only briefly introduced here to provide the necessary information for simulation. The quartz tube reactor of 6 mm inner diameter was positioned in the centerline of the heating furnace, with an effective heating length of 460 mm. A flame ionization detector (GC-FID, GC2014 SHIMAZU) and a gas chromatography with a thermal conductivity detector (GCTCD, GC2014 SHIMAZU) were used to detecting species of below C₅.

Three equivalence ratios of 0.5, 1.0 and 1.5, covering fuel-lean to fuel-rich conditions, were adopted in the experiments at atmospheric pressure in the temperature range of 600–1100 K. The total flow rate maintained constant at 2 standard-l/min. High purity argon was used as balance gas. The test mixtures are shown in Table 3.

2.3. Computational approach

The simulation work in the present study was conducted using Chemkin-Pro-software. For the simulation of IDT measurements,

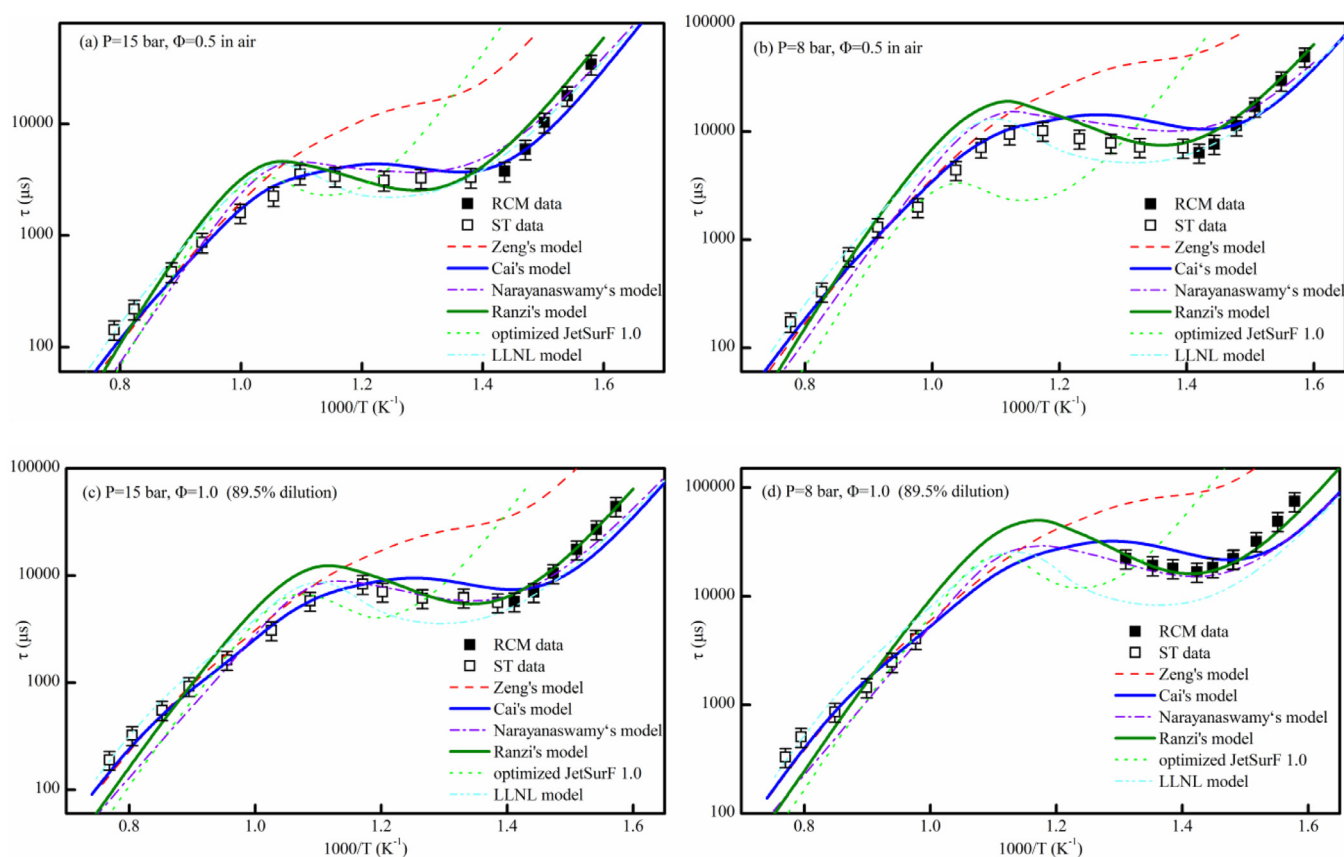


Fig. 3. Comparison of the present experimental data with the predictions from existing kinetic models [25,26,39,41,42,44].

the Closed Homogeneous Batch Reactor model was adopted with the constant Volume (Const.) constraint. In order to mimic the facility effect, additional RCM simulations using the volume histories derived from the non-reactive runs were performed for RCM measurements. For flow reactor oxidation, the simulation was performed using the Plug Flow Reactor model.

3. Kinetic model development

3.1. The need for an improved model

IDTs are one of the kinetic targets that models should be able to predict. The present experimental data was firstly used to evaluate the existing models, including Ranzi's model [44], optimized JetSurF 1.0 model [26], LLNL model [39], Narayanaswamy's model [42], Cai's model [41] and Zeng's model [25]. Figure 3 shows a comparison between the simulations using the above models and experimental data under several conditions. It is found that there is great discrepancy between the experimental data and the simulations from the optimized JetSurF 1.0 model, especially at lower temperatures. LLNL model and Narayanaswamy's model capture the trend of IDTs with the variation of temperature; however, improvements are still needed to improve the performance of the two models. Among these models, Ranzi's model provides the best predictions in low-to-intermediate temperature range, but it captures neither the slope nor the magnitudes of IDTs at a satisfactory level at higher temperatures. Zeng's model well follows the trend of IDTs in high-temperature range; regrettably, it overpredicts the IDTs in low-to-intermediate temperature range. Considering the entire temperature range, Cai's model provides a better prediction overall compared with the mentioned models, although there is still a need to improve its performance in low-to-intermediate tempera-

ture range. These discrepancies with the present experimental data again call for an improved kinetic model for n-dodecane.

3.2. Development of the kinetic model

In order to facilitate the merging of mechanisms of components into a merged mechanism for surrogate fuels, it will be better to develop mechanisms in a modular manner for individual component. The proposed model was composed of three modules, namely base model, high temperature reaction scheme and low temperature scheme.

In the present model, the widely used AromcoMech 2.0 model [58] was chosen as the base model. The high temperature pathways were developed based on the reaction classes proposed by Sarathy et al. [39]. Since Zeng's model presented a good performance in predicting the high temperature reactivity of n-dodecane and had been validated against experimental data comprehensively (as shown in literature [25]), the rate constants in the present high temperature reaction scheme were mainly taken from this model, except those for the isomerization of alkyl radicals and H abstraction reactions, which were taken from Cai's model.

Since the low temperature reaction chemistry is of particular interest in the present study, it would be best to adopt Cai's model [41], which has included these reaction schemes and provided relatively better predictions, as the reference mechanism. The low temperature pathway was basically extracted from Cai's model, but some modifications were made based on the recent kinetic studies to improve the prediction of model in low temperature range. The details of the modifications will be described specifically as follows. As the first step of the low temperature chain branching sequence, the rate coefficients for O_2 addition to alkyl radicals were prescribed according to the rate rules proposed by Miyoshi [59].

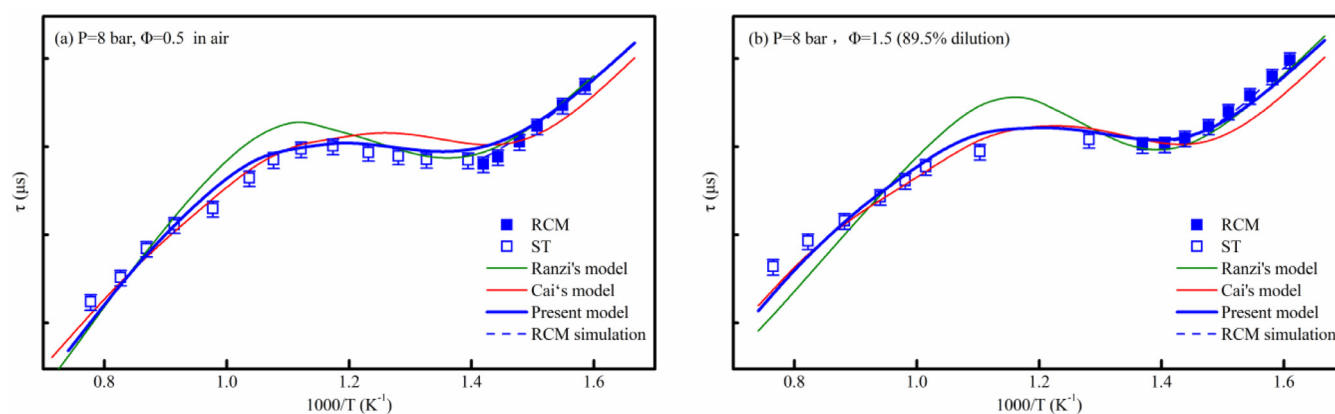


Fig. 4. Comparison of the predictions from the present model and the existing models [41,44].

The same rate rules were used for the second O_2 addition. The isomerization of peroxy alkylhydroperoxide radicals (OOQOOH) to form Ketohydroperoxide, together with the subsequent decomposition of ketohydroperoxide finalize the low temperature chain branching process, and are crucial for the low temperature reactivity of fuels. In this study, rate coefficients for the decomposition of peroxy alkylhydroperoxide radicals were prescribed based on the rate rule by Sharma et al. [60]. While for the subsequent decomposition of ketohydroperoxide via O-O scission, the rate coefficients were assigned depending on carbon type. For primary carbon, a frequency (A) factor of $1.5 \times 10^{16} \text{ s}^{-1}$ and an activation energy of $40.8 \text{ kcal mol}^{-1}$ were assigned; for secondary carbon, a frequency factor of $1.05 \times 10^{16} \text{ s}^{-1}$ and an activation energy of $42.3 \text{ kcal mol}^{-1}$ were assigned. Concerted elimination and the formation of cyclic ether were key reaction responsible for the NTC behavior of large hydrocarbons. In this study, the concerted elimination was prescribed using the rate rules proposed by Zhang et al. [61] for n -alkanes; while the cyclic ether formation was assigned using the rate rules proposed by Villano et al. [62].

Figure 4 shows a comparison between the predictions from the relatively better existing models and the present updated model. It is found that the present model follows the reactivities of the combustible mixtures better than the existing model, indicating the improved ability of the model in describing the combustion chemistry of n -dodecane.

4. Results and discussion

4.1. Autoignition characteristics of n -dodecane under various conditions and model simulation

The autoignition delay times of n -dodecane/ O_2/N_2 mixture were studied over wide conditions to cover various pressures, equivalence ratios, and dilution ratios. Figures 5–8 shows the effects of different parameters on the IDTs of n -dodecane. In order to facilitate the evaluation of the present model, the Const. simulations using constant volume constraint and RCM simulations for RCM data were displayed as well.

Figure 5 shows the influence of pressure on reactivity of n -dodecane by comparing IDTs for this target fuel at different pressures. Overall, the data sets obtained from RCM and ST appear consistent with each other. This is distinct from the case for high octane number fuels, e.g. *iso*-octane [63] and *iso*-dodecane [64], for which there is a 2–3 factor of discrepancy between the data obtained from the two types facilities. Such a discrepancy is considered to be attributed to the boundary effects and other non-ideal conditions [63,65], such as unexpected breakage of diaphragms in ST measurement and heat loss in RCM measurements.

These non-ideal effects become more prominent with increasing measuring time scale, therefore, the consistency for the present fuel of high-reactivity is expected due to its much shorter IDTs under the same conditions. Under all conditions, a typical NTC behavior is observed. It is found that shorter IDTs were obtained at higher pressures for all the presented combustible mixture, which were expected since the increasing reactant concentration enhances the reaction rate. Furthermore, it is evident that the effect of pressure is most pronounced in the NTC regime. This trend is extensively observed in the autoignition of hydrocarbons [47,64–67], and is believed to be the consequence of the increasing significance of pressure-dependent reactions (e.g. $\text{H}_2\text{O}_2(+\text{M})=\text{OH}+\text{OH}(+\text{M})$) in this temperature range [64]. It is found that there is excellent agreement between the experimental data and simulations, almost spanning all temperatures, pressures, equivalence ratios and dilution ratios.

Figure 6 presents the IDTs of n -dodecane at different equivalence ratios. Since the oxygen molar fraction for the shown mixtures were almost kept constant (around 10.44%), the effect of equivalence ratio means the effect of fuel molar fraction. Due to the excessively lean fuel in the mixture of equivalence ratio of 0.5, the mixture cannot ignite at low temperatures and thereby RCM measurements were not performed for this mixture. Overall, the model well captures the variation trend of IDTs and the effect of fuel molar fraction. From the simulations it is found that the effect of fuel molar fraction is more prominent in the fuel-lean condition compared with fuel-rich condition. At the high temperature region, the effect of fuel molar fraction is much weaker compared with NTC region. Above 1200 K, there is a crossover point between the mixtures of different equivalence ratios, implying that the reactivity of the fuel-lean mixture begins to exceed that of fuel-rich mixture. The occurrence of the crossover at high temperature is due to the competition between the increasingly significant reaction $\text{O}_2+\text{H}=\text{O}+\text{OH}$ and the H abstraction from fuel by H radicals. Since the reaction $\text{O}_2+\text{H}=\text{O}+\text{OH}$ control the overall reactivity in high temperature range, the higher fuel molar fractions in fuel-rich mixtures would enhance this competition and thereby render the decreased reactivity. This behavior is well captured by the model.

The effect of oxygen molar fraction is presented in Fig. 7. As expected, the increase of oxygen concentration promotes the autoignition of mixtures. It's worth noting that there is no crossover point observed in high temperature, which is different from the effect of fuel molar fraction. As stated above, the dominant chain branching reaction at high temperatures is $\text{O}_2+\text{H}=\text{O}+\text{OH}$, the increase of oxygen (fuel maintain constant) naturally would increase the overall reactivity. Another interesting phenomenon is that under some conditions the RCM simulations are shorter than the Const. simulations. For the RCM simulations in the present study,

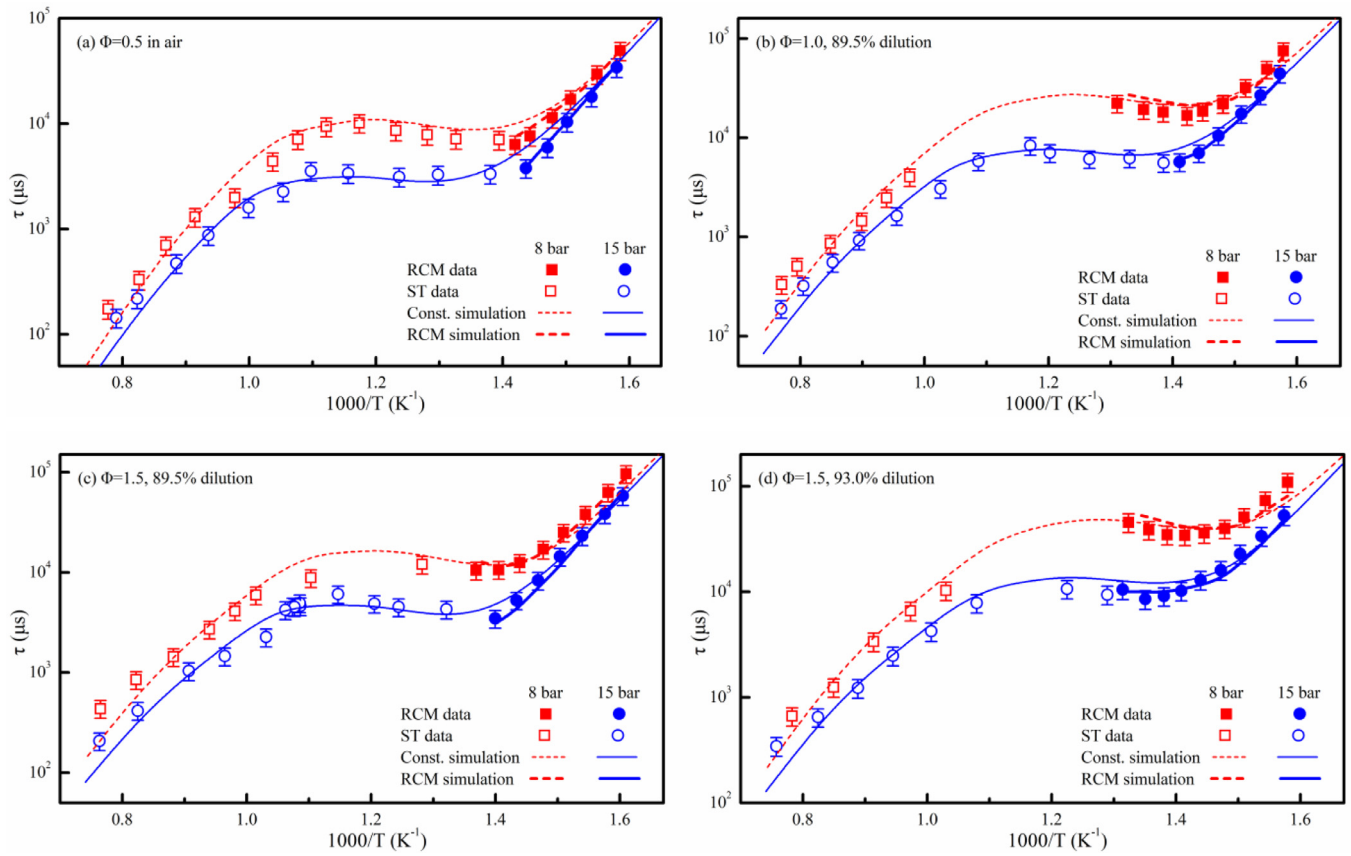


Fig. 5. Effect of pressure on autoignition characteristics of n-dodecane.

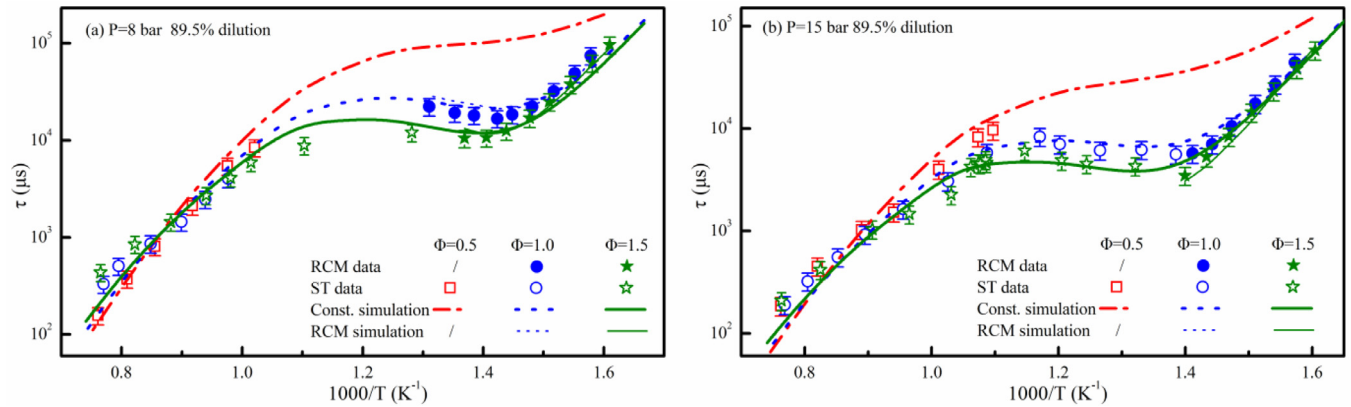


Fig. 6. Effect of equivalence ratio on autoignition characteristics of n-dodecane.

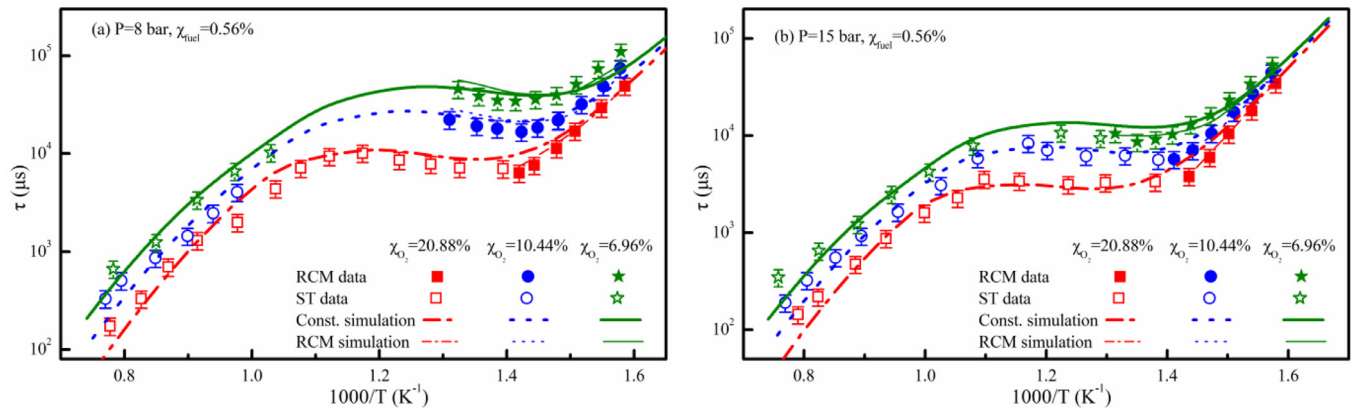


Fig. 7. Effect of oxygen mole fraction on autoignition characteristics of n-dodecane.

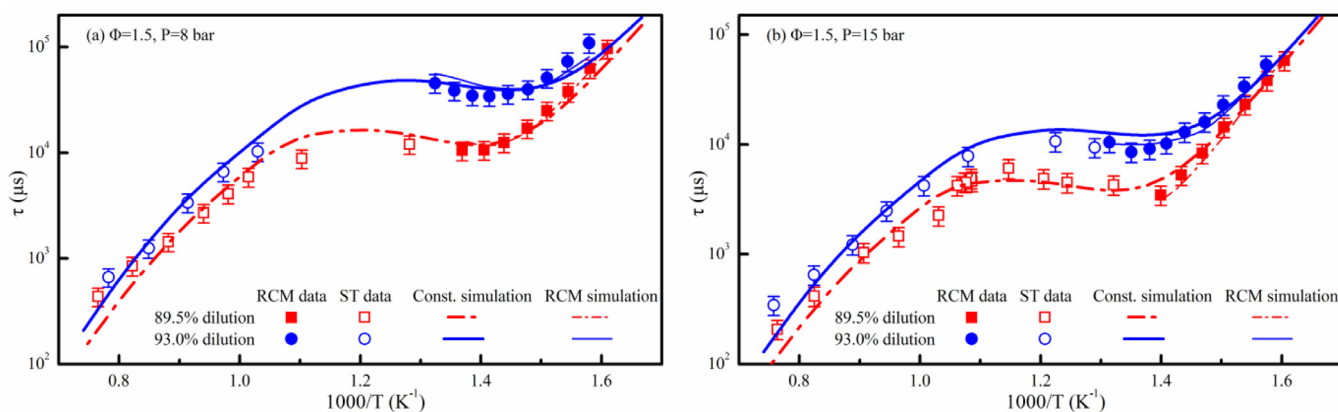


Fig. 8. Effect of dilution ratio on autoignition characteristics of n-dodecane.

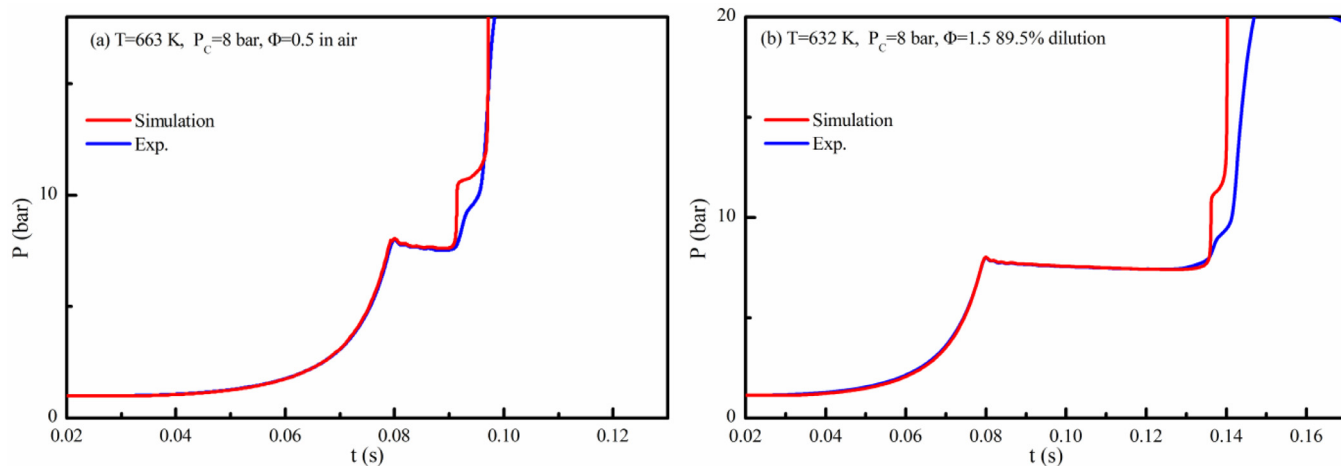


Fig. 9. Comparison of measured and predicted pressure profiles at low temperatures.

the compression stroke is also included in the simulation to account for its effect on autoignition. According to Sung and Curran [53], aside from the heat loss, the active radicals formed during the compression stroke would affect the autoignition as well, i.e., promoting the subsequent reactions. Therefore, whether Const. simulations are shorter than RCM simulations depends on the combined effect of the two factors.

Figure 8 presents a comparison between the reactivity of mixtures with the same equivalence ratio but different dilution ratio. The increase of dilution ratio decreases the overall reactivity notably. The effect of dilution is considered to be attributed to the variation of reactant concentration and the thermal effect of the balance gas. Generally, the model satisfactorily captures the effect of dilution ratio.

Figure 9 further compares the measured pressure traces with the predicted ones using RCM simulation at low temperatures. It is found that the model well follows the reactivity of the test fuel. The computational pressure traces almost completely overlap with experimental ones, implying that the use of volume histories derived from non-reactive runs to mimic the facility effect is reasonable. Both the predicted first stage and overall IDTs are in good agreement with the experimental ones, although the predicted pressure rise after the first stage ignition is slightly higher than the experimental value.

4.2. Speciation in the flow reactor oxidation

The oxidation of n-dodecane was conducted as a function of the initial reaction temperature at 1atm using the flow reactor at

SJTU. In order to avoid thermal runaway due to the heat release of oxidation, the combustible mixtures were highly diluted using high-molar-fraction balance gas argon to approximate the isothermal condition. Throughout the experiments, the mole fraction of n-dodecane was maintained 0.3% and the equivalence ratio was varied by altering the oxygen molar fraction.

Figures 10–12 shows the speciation profiles of several key intermediate species during n-dodecane oxidation for equivalence ratios of 0.5, 1.0 and 1.5, respectively. Generally, good agreement was found between the experimental results and the simulations from fuel-lean to fuel-rich conditions, although the model underestimated the concentrations of C_3H_8 . There is a slight drop in the O_2 profiles around 725–775 K (depend on equivalence ratio), suggesting the weak and very narrow temperature window of low temperature reactivity. CO is considered to be a good indicator of low temperature reactivity. Corresponding to the O_2 profiles, there is a peak in the CO profiles in low temperature range, implying the NTC behavior of this fuel. Furthermore, the low temperature reactivity decreases with the increase of equivalence ratio. For the other species, there is no noticeable or detectable variation of concentration in the low temperature range. However, the situation is distinct in the experiment of Kurman et al. [27], in which n-dodecane shows a much stronger low temperature reactivity. In addition to the short residence time, the relatively low pressure (much lower than 8 atm adopted in Kurman's measurements) is considered to be the major cause of the weak low temperature reactivity in the present study. Above 900 K, O_2 starts to decay quickly, indicating that the high temperature pathway begins to act. The major final product CO_2 is mainly generated at high temperatures

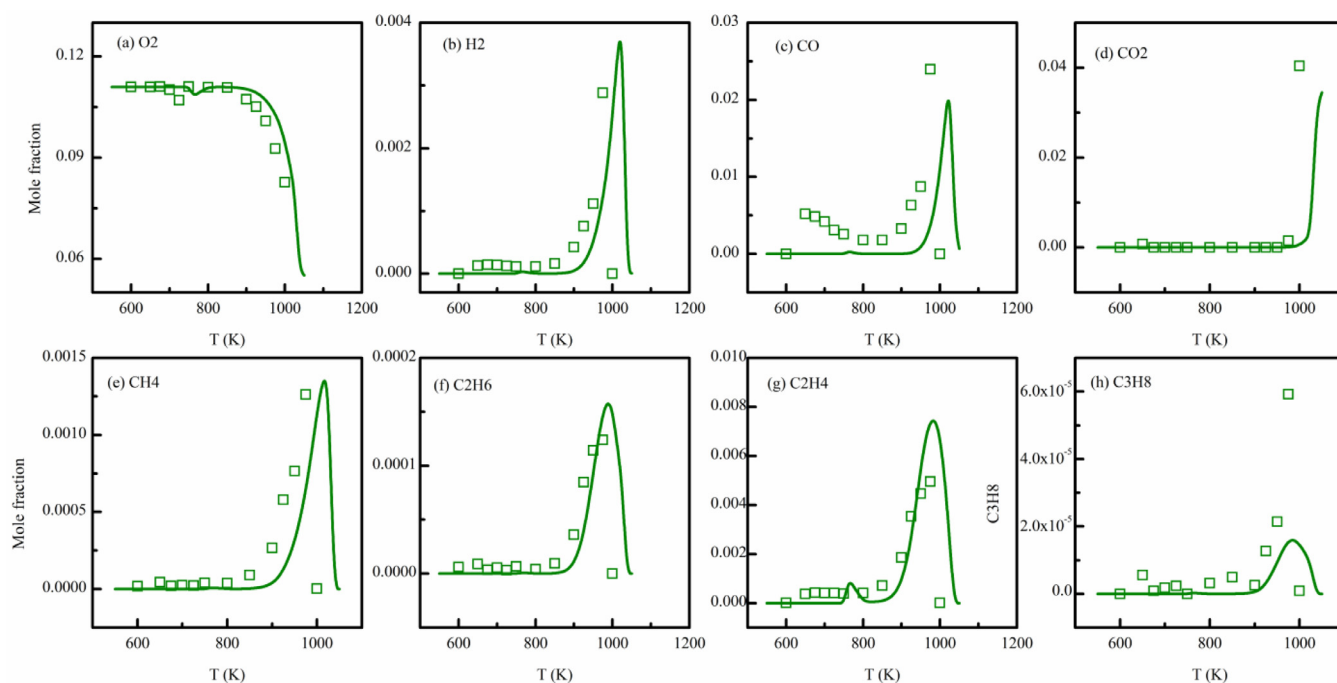


Fig. 10. Oxidation speciation profiles of n-dodecane under condition of atmospheric pressure and $\Phi = 0.5$. Symbols are experimental data and lines are simulations.

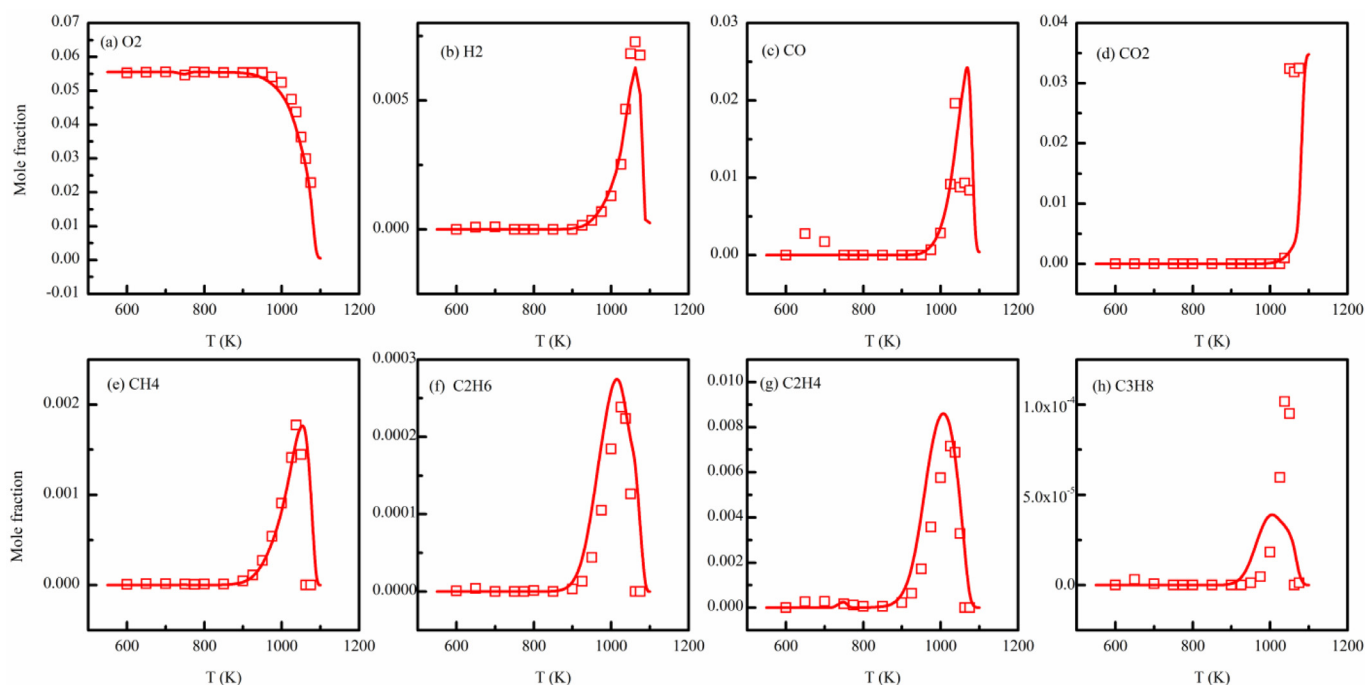


Fig. 11. Oxidation speciation profiles of n-dodecane under condition of atmospheric pressure and $\Phi = 1.0$. Symbols are experimental data and lines are simulations.

and just peak after CO. By comparing Figs. 10–12, it is found that the formation of CO and CO_2 is more favored under fuel-lean conditions, because the higher oxygen concentration tends to promote the generation of these oxygenated products. While the other stable intermediate species, including small alkanes and alkenes, are more favored under fuel-rich conditions. The formation of alkanes usually involves chain termination reactions, e.g., two CH_3 radicals recombine to produce ethane, and thereby suppresses the overall reactivity of the system. This accounts for the much steeper slope of oxygen profile under fuel-lean conditions. Among the alkane products, methane is most abundant partly due to its thermal sta-

bility and low reactivity. Ethene is also formed in a considerable level even higher than all alkanes. In addition to its good thermal stability, the multiple formation pathway of C_2H_4 also contributes to its accumulation, for instance, following the unimolecular decomposition of the parent fuel, the derived radicals (e.g., C_4H_9 , C_3H_7 , C_2H_5 etc.) can easily decompose to produce C_2H_4 via beta scission or H elimination.

There are slight temperature shifts in the profiles between the different equivalence ratios. This behavior is well captured by the model. Note that there is a 25 K phase shift between the simulations and experimental data for equivalence ratio of 0.5. This may

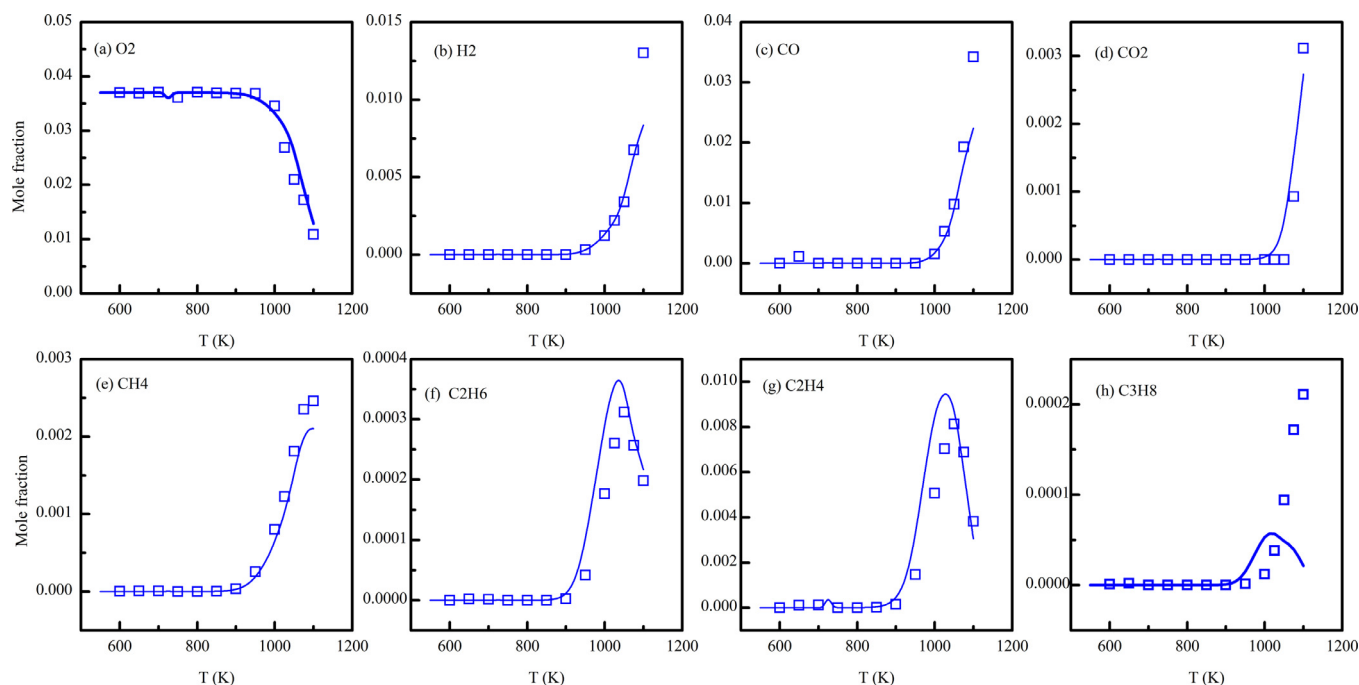


Fig. 12. Oxidation speciation profiles of n-dodecane under condition of atmospheric pressure and $\Phi = 1.5$. Symbols are experimental data and lines are simulations.

be attributed to the higher oxygen molar fraction in this mixture. Due to the higher oxygen concentration, the actual temperature rise caused by the heat release during oxidation for this mixture would be larger than the other two mixtures. This could be confirmed by comparing the different yields of CO_2 in Figs. 10–12, as reaction $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ is one of the exothermic reactions to achieve the majority of heat release. Consequently, the temperature rise for mixture of equivalence ratio of 0.5 could not be neglected and the actual reaction temperature would be higher than the nominal temperature. Therefore, the reactions tend to occur at a lower nominal temperature. Considering these non-ideal effects in experimental measurements, the predictions are in excellent agreement with the experimental data.

5. Validation of the present model against literature data

To comprehensively evaluate the performance of the model, we use the model to simulate the available experimental data in literature. Here, the databases adopted for evaluation mainly focus on oxidation environment while the pyrolysis data was not considered.

5.1. Ignition delay times in literature

The autoignition characteristics of n-dodecane has been studied in shock tubes previously and mainly focus on intermediate-to high temperatures, including the work of Vasu et al. [17], Shao et al. [20], Shen et al. [19] and Davidson et al. [18]. These experimental data were all used to evaluate the present model, as shown in Fig. 13. Among these data, Vasu's data and Shen's data were scaled using the correlation of $\tau - P^{-1}$ provided in their studies; since the equivalence ratio in Shao's experimental showed a big fluctuation, this dataset was simulated using the exact conditions for each experiment. The model does well in predicting the high temperature reactivity, including both the magnitudes and slopes of IDT profiles. It also captures the NTC temperature window, although slightly underpredicts the IDTs in this region. The model reproduces the rather limited low temperature data excellently

as well. There is a relatively large discrepancy between the data provided by Davidson et al. [18] and simulations above 1250 K. Given that this dataset was obtained in an aerosol ST, there may be a large uncertainty due to the adsorption and deposition of fuel, which has been reflected by the discreteness of these data. Therefore, such a discrepancy is understandable. Overall, the model reproduces the autoignition behaviors of n-dodecane well under these conditions.

5.2. Speciation of shock tube oxidation

The oxidation and pyrolysis of n-dodecane was investigated by Malewicki and Brezinsky [23] in a single pulse heated high-pressure shock tube. The experiments cover a wide range of 867–1739 K, and equivalence ratios of 0.46–2.05. The experimental data spans pressures from 19 to 74 atm, which provides the opportunity to test the performance of the model at very high pressures. In this section, simulations were performed using the Closed Homogenous Batch Reactor model in Chemkin-Pro-software at the exact fuel/ O_2 /argon compositions. Figures 14 and 15 present a comparison of the measurements and simulations under fuel-lean and fuel-rich conditions, respectively. Under the both conditions, it is found in Figs. 14(a) and 15(a) that the consumption of oxygen began after almost the complete decomposition of fuel, implying that the fuel was consumed through other pathways (mainly H abstraction according to ROP analysis) before oxidation. This model accurately captures the decay of the two reactants. The model also does well in predicting the variation trends of major oxygenated products and small alkanes, which are the major products during the oxidation. The 1-alkenes intermediates from C_2 to C_{11} were important products and their concentrations were measured as a function of temperature as well. These species were mainly formed via the decomposition of radicals resulting from unimolecular decomposition of the fuel. Generally, the smaller alkenes peaked after the larger ones, as the decay of larger alkenes through pathways such as retro-ene decomposition continued to contribute to the formation of smaller ones. The model follows this trend accurately, as shown in panels (b) and (c) in Figs. 14 and 15. Furthermore,

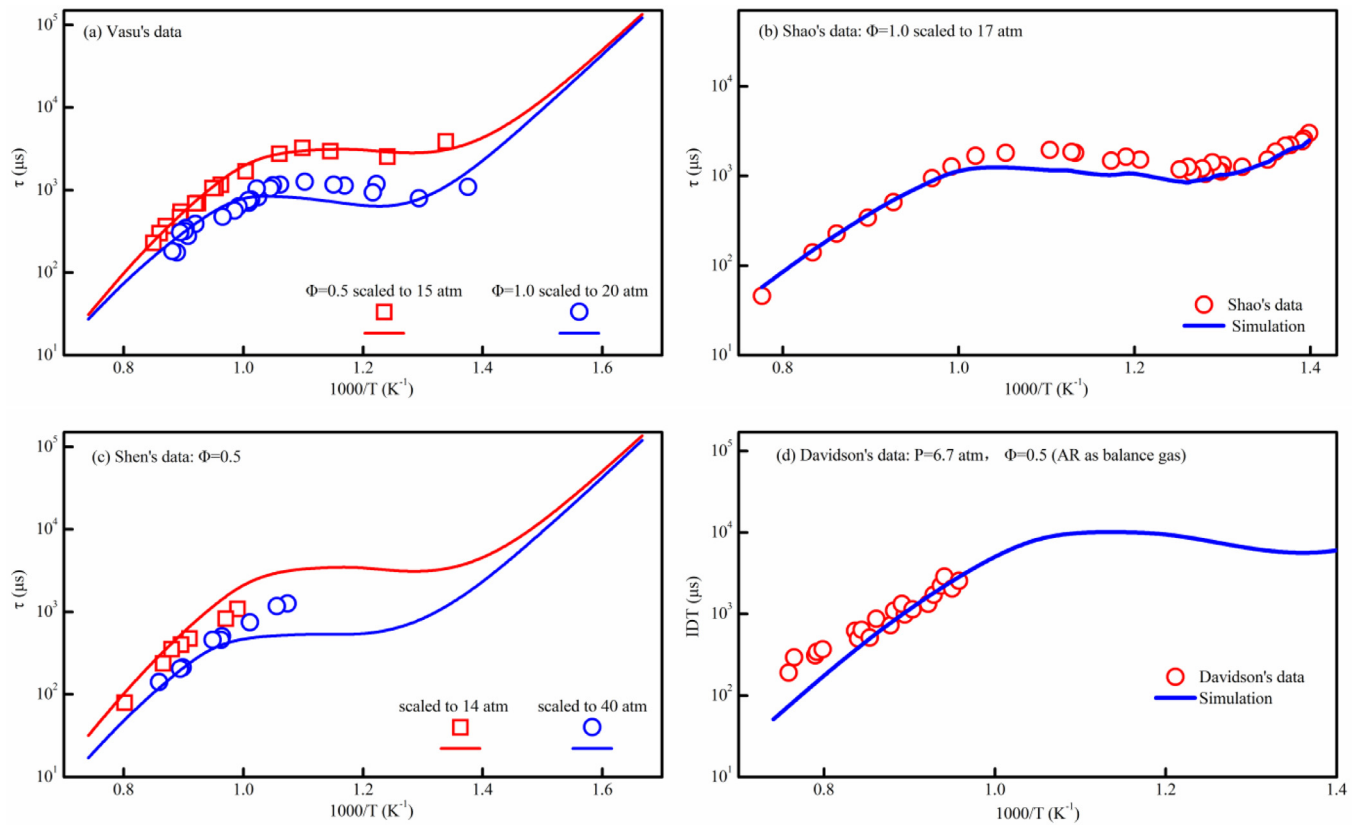


Fig. 13. Ignition delay times of n-dodecane/air mixtures. Symbols are experimental data in literature [17–20], and lines are simulation using the present model.

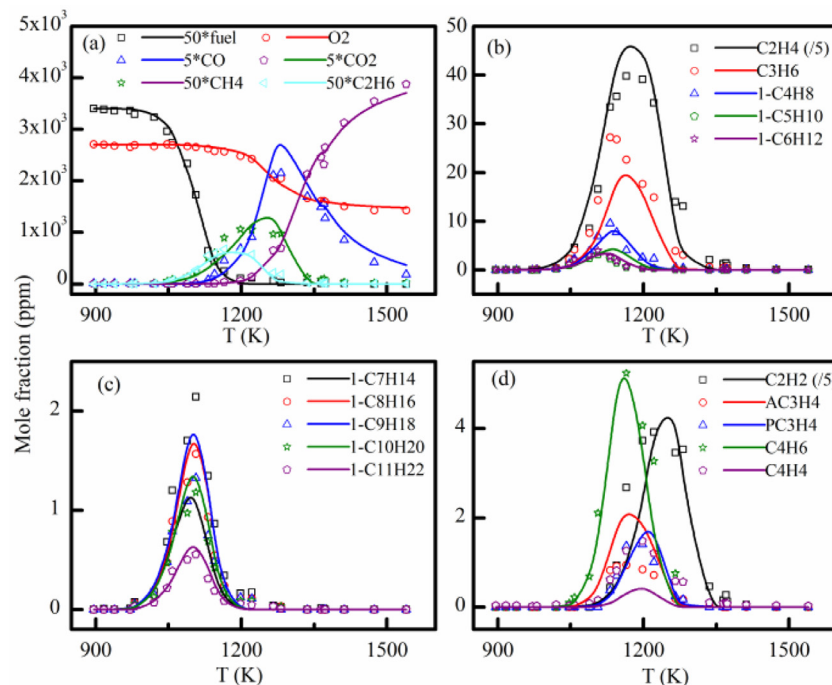


Fig. 14. Molar fraction of major stable species during the oxidation of n-dodecane at $\Phi=0.46$ and nominal pressure of 24.45 atm. Symbols are experimental data [23] and lines are simulations.

the model provides predictions for these species with nearly the same magnitude with experimental results, except for underpredicting 1-C₃H₆ and 1-C₇H₁₄ by approximate 30%. Alkynes, typically considered as the precursors of pollutants, were also simulated in Figs 14(d) and 15(d). The model presents a very good performance

against the data under fuel-lean condition. Under fuel-rich condition, the model slightly underestimates these unsaturated hydrocarbons, implying further improvements are needed for the model. Overall, the model shows its excellent ability in reproducing the speciation profiles of n-dodecane oxidation in these conditions.

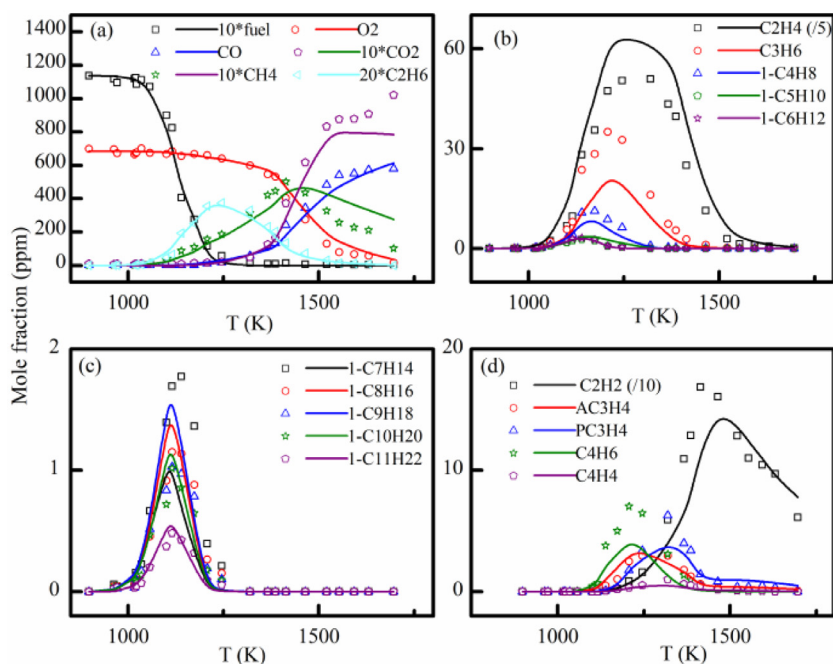


Fig. 15. Molar fraction of major stable species during the oxidation of n-dodecane at $\Phi=2.05$ and nominal pressure of 49.55 atm. Symbols are experimental data [23] and lines are simulations.

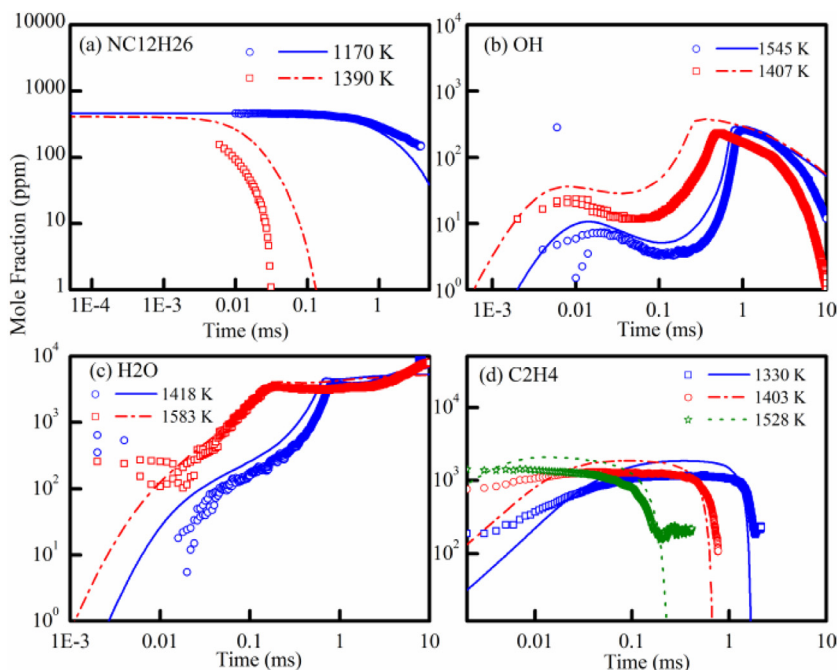
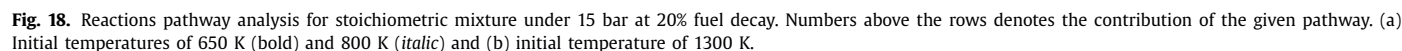
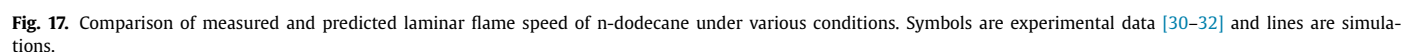


Fig. 16. Time-history speciation profiles during n-dodecane oxidation in a shock tube. Symbols are experimental data [21] and lines are simulations.

5.3. Species time histories in shock tube oxidation

Measurements of the time-histories of crucial species are essential kinetic target in evaluating kinetic models and provide insight into the reaction process from the microscopic perspective different from global parameters. Davidson et al. [21] measured the concentration time-histories of several key species behind reflected shock waves. The experiments were carried out in the temperature range of 1300–1600 K for diluted stoichiometric n-dodecane/ O_2 /Ar mixtures at around 2 atm. Figure 16 presents the simulation results using the Closed Homogenous Batch Reactor model in Chemkin-

Pro-software at the exact conditions. In Fig. 16(a), the model captures the trend of the fuel depletion, although difference still exists at 1390 K. The evolution of OH radical, one of the most significant radicals in the combustion process, was simulated and shown in Fig. 16(b). The good agreement of the predicted value with the experimental data suggests that the prescribed rate coefficient for H abstraction from fuel by OH radical is appropriate, which is the major channel for OH depletion. In Fig. 16(c), the predicted concentrations of H_2O were in good agreement with the experimental results, including the peaks and plateaus. The agreement of the C_2H_4 yields with the predictions in Fig. 16(d) indicates the branching



ratio pathway of n-dodecane decomposition to form ethene is basically reliable [21]. Generally, the model well reproduces the measurement results.

5.4. Laminar flame speed

Available laminar flame speed measurements for n-dodecane include the work of Hui et al. [31], Kumar et al. [30] and Ji et al. [32]. These data were all used to assess the present model. Simulations were conducted using the Premixed Laminar Flame-Speed Calculation module in Chemkin-Pro-software. The results are illustrated in Fig. 17. In Fig. 17(a), the model accurately follows the dependence of laminar flame speed on pressure, and the simulation results show excellent agreement with Hui's data in general. In Fig. 17(b), the computational values are obviously lower than Kumar's data at fuel-rich conditions. However, compared with another dataset measured by Ji et al. at the very similar conditions, as well as Hui's data at 1 atm, Kumar's data seems to be higher at the fuel-rich side. It is further found in Fig. 17(b) that the prediction from the model is almost exactly the same as the experimental data of Ji et al. [32]. Considering the discrepancy between Kumar's data and the other two datasets, it can be deemed that the model can provide reliable predictions for laminar flame speed at these conditions. Of course, additional data are still needed to support this remark.

6. Reaction pathway analysis

The good performance of the model in reproducing the experimental data motivates the interest to perform a detailed kinetic analysis using the mechanism for a better understanding of the combustion chemistry of n-dodecane. In this section, reaction pathway analysis was performed for 15 bar at three temperatures of 650, 800 and 1300 K to cover low, intermediate and high temperatures for stoichiometric temperatures. For the other mixtures and pressures conditions, the branching ratios for each channel are only slightly changed, thus the results illustrated in Fig. 18 is representative.

At all three temperatures, the overwhelming majority of fuel depletion occurs via H abstraction reactions. Of the abstractors, OH radicals contribute the most. However, radicals HO_2 and H play an increasing significant role with the increasing temperature, and at 1300 K H radicals almost consume equivalent fuel compared with OH. On the other hand, it is well established that reaction $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$ is the chain branching reaction governing the overall reactivity at high temperatures. Therefore, the increasing fuel molar fraction would render H abstraction from parent fuel to compete with reaction $\text{H} + \text{O}_2 \rightarrow \text{OH} + \text{O}$ for H radicals more effectively, leading to inhibited reactivity. This accounts for the crossover point in the high temperature region in Fig. 6. As one of the initial reaction classes, unimolecular decomposition contributes little to fuel decay at low-to-intermediate temperatures, but this channel become relatively important at high temperatures. For instance, it accounts for 12.40% of fuel conversion at 1300 K. At 650 and 800 K, the fuel derived radicals mainly undergo O_2 addition and isomerize to QOOH species. As temperature increases, the contributions of these low temperature reaction pathway decrease and the beta-scission channel become increasing importance. As temperature reaches 1300 K, the fuel derived radicals are primarily consumed via C–C bonds dissociation, as shown in Fig. 18(b). Due to the diversity of QOOH species, only one γ -QOOH species was shown for each fuel derived radicals for simplicity. It is found in Fig. 18(a) that over 97% of these species undergo the second O_2 addition at 650 K to achieve the low temperature degenerate chain branching sequence. However, this proportion decreases to less than 67% at 800 K, and the rest γ -QOOH species undergo

cyclic formation or decomposition reactions. These reaction channels tend to produce low reactivity cyclic ethers and alkenes, and thereby decrease the overall reactivity, leading to the NTC behavior of large hydrocarbon in intermediate temperature range.

7. Conclusion

The autoignition of n-dodecane, which was seldom investigated at low temperatures using RCM previously, was studied over low-to-high temperature ranges by combining a heated RCM and ST in the present study. The experimental conditions cover a wide range of temperature (621–1320 K), pressure (8 and 15 bar), equivalence ratio (0.5, 1.0 and 1.5) and dilution ratio (79%, 89.5% and 93%). The effect of these operating parameters on autoignition characteristics of n-dodecane, were comprehensively presented. Furthermore, the oxidation of n-dodecane was investigated in a flow reactor for equivalence ratio of 0.5, 1.0 and 1.5 over the temperature 600–1100 K under atmospheric pressure. Several crucial stable products, including O_2 , H_2 , CO , CO_2 , CH_4 etc., were detected and their speciation profiles were obtained as a function of temperature. An improved kinetic model was developed in this study based on the kinetic model for alkanes developed by Cai et al. [41]. The model was evaluated against the present experimental data and a wide range of datasets in literature. Good agreements were observed between predictions from the model and experimental data. The present model is anticipated to be used as sub-mechanism for the development of kinetic model for multi-component jet fuel or diesel surrogate.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.11.014.

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